CHAPTER 2:
Atomic Structure and Interatomic Bonding

ISSUES TO ADDRESS...

- What promotes bonding?
- What types of bonds are there?
- What properties are inferred from bonding?

Atomic Structure (Freshman Chem.)

- atom – electrons – 9.11×10^{-31} kg
  protons
  neutrons

- atomic number = # of protons in nucleus of atom
  = # of electrons of neutral species

- A [=] atomic mass unit = amu = 1/12 mass of ^{12}C

Atomic wt = wt of 6.022×10^{23} molecules or atoms

1 amu/atom = 1g/mol

C	12.011
H	1.008 etc.

AVAGADRO'S NUMBER = 6.022×10^{23} = N_A

ATOMIC OR MOLECULAR WEIGHT = N_A x WEIGHT PER ATOM.

number of neutrons = N

number of protons = Z

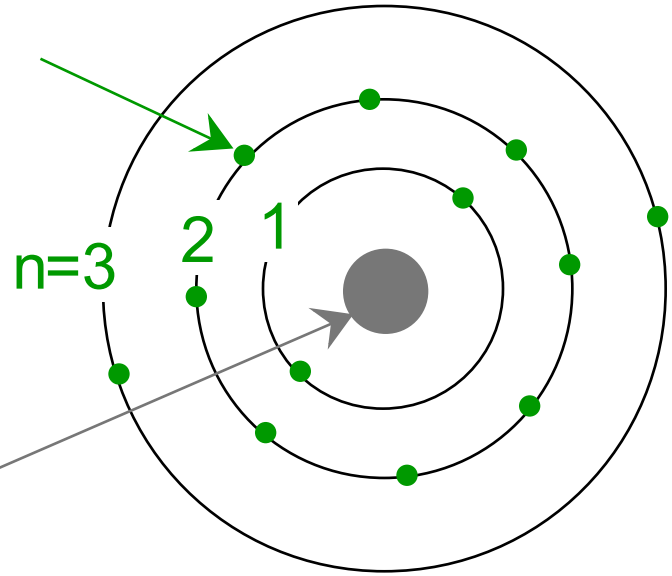
$$A = Z + N \quad (2.1)$$

Atomic Structure

- Valence electrons determine all of the following properties
    1) Chemical
    2) Electrical
    3) Thermal
    4) Optical

BOHR ATOM

orbital electrons:
n = principal
quantum number



Adapted from Fig. 2.1, *Callister 6e.*

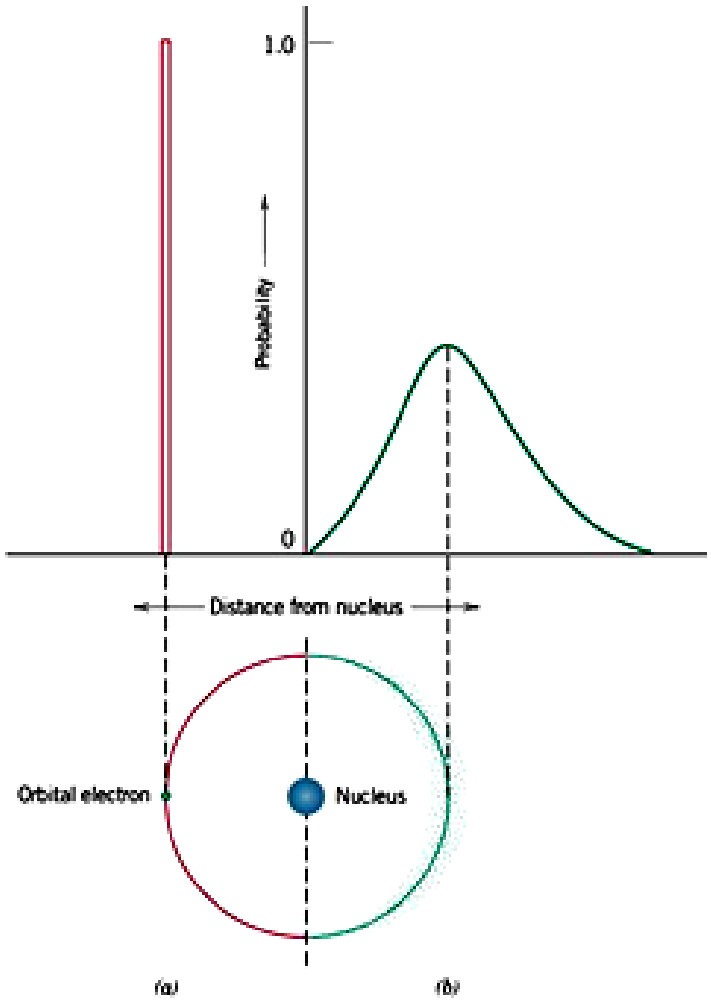
Nucleus: Z = # protons

= 1 for hydrogen to 94 for plutonium

N = # neutrons

Atomic mass A ≈ Z + N

WAVE MECHANICAL MODEL OF ATOM



Electronic Structure

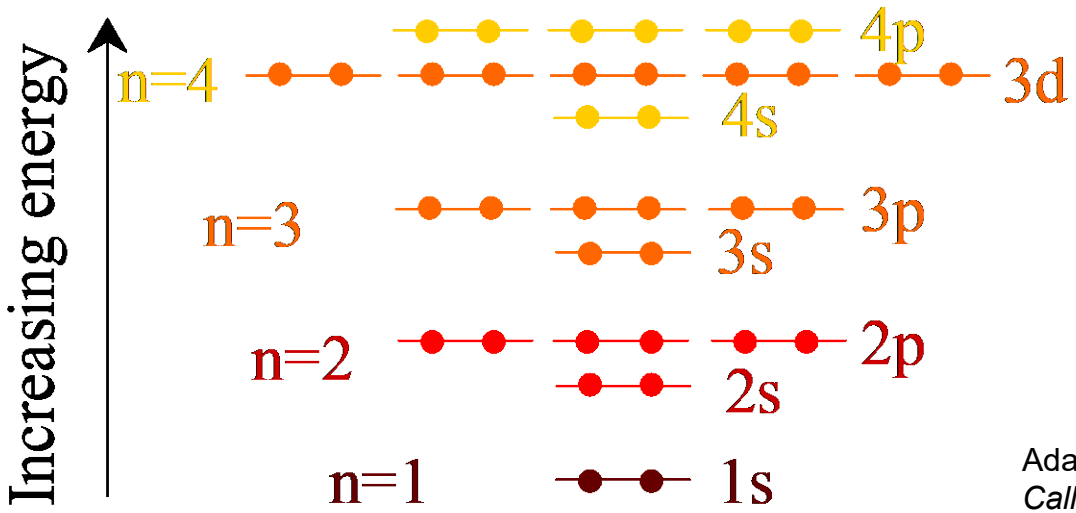
- Electrons have wavelike and particulate properties.
  - This means that electrons are in orbitals defined by a probability.
  - Each orbital at discrete energy level is determined by quantum numbers.

Quantum #	Designation
n = principal (energy level-shell)	K, L, M, N, O (1, 2, 3, etc.)
l = subsidiary (orbitals)	s, p, d, f (0, 1, 2, 3,…, n-1)
m_l = magnetic	1, 3, 5, 7 ($-l$ to $+l$)
m_s = spin	½, -½

ELECTRON ENERGY STATES

Electrons...

- have discrete energy states
- tend to occupy lowest available energy state.



Adapted from Fig. 2.5, *Callister 6e.*

STABLE ELECTRON CONFIGURATIONS

Stable electron configurations...

- have complete s and p subshells
- tend to be unreactive.

Z	Element	Configuration
2	He	$1s^2$
10	Ne	$1s^2 2s^2 2p^6$
18	Ar	$1s^2 2s^2 2p^6 3s^2 3p^6$
36	Kr	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6$

Adapted from Table 2.2, *Callister 6e.*

SURVEY OF ELEMENTS

- Most elements: Electron configuration not stable.

Element	Atomic #	Electron configuration	
Hydrogen	1	$1s^1$	
Helium	2	$1s^2$	(stable)
Lithium	3	$1s^2 2s^1$	
Beryllium	4	$1s^2 2s^2$	
Boron	5	$1s^2 2s^2 2p^1$	
Carbon	6	$1s^2 2s^2 2p^2$	
…		…	
Neon	10	$1s^2 2s^2 2p^6$	(stable)
Sodium	11	$1s^2 2s^2 2p^6 3s^1$	
Magnesium	12	$1s^2 2s^2 2p^6 3s^2$	
Aluminum	13	$1s^2 2s^2 2p^6 3s^2 3p^1$	
…		…	
Argon	18	$1s^2 2s^2 2p^6 3s^2 3p^6$	(stable)
…	…	…	
Krypton	36	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4^6$	(stable)

Adapted from Table 2.2, *Callister 6e.*

- Why? Valence (outer) shell usually not filled completely.

Electron Configurations

- Valence electrons – those in unfilled shells
- Filled shells more stable
- Valence electrons are most available for bonding and tend to control the chemical properties

  – example: C (atomic number = 6)

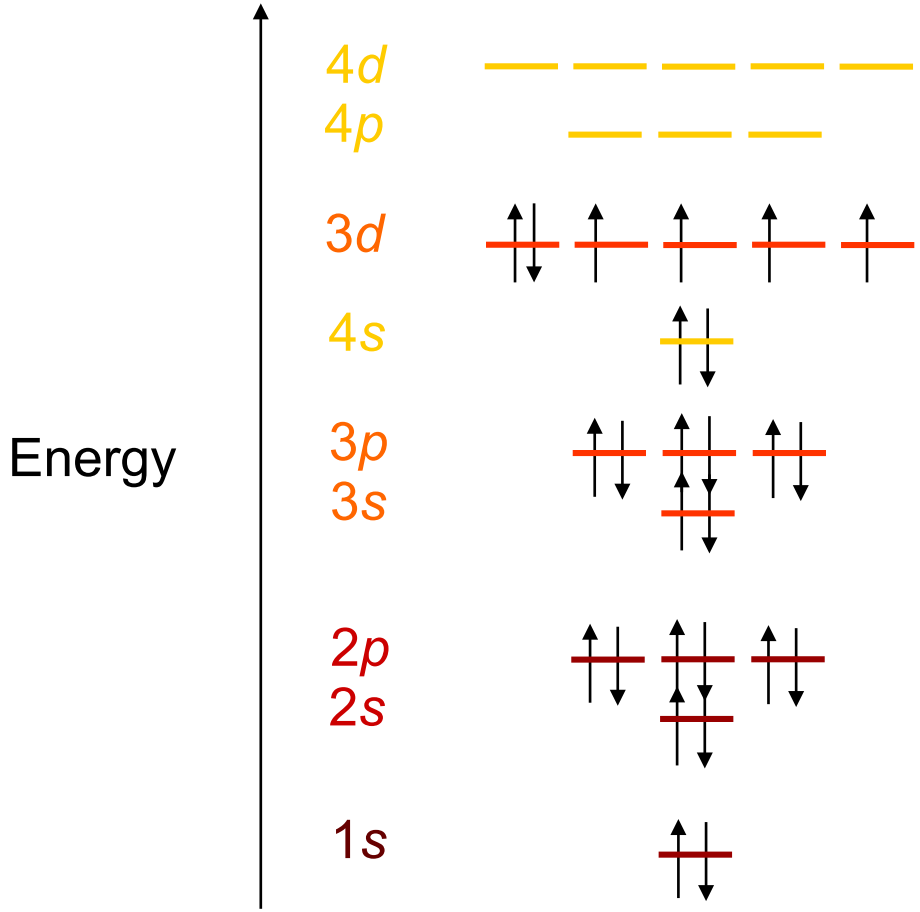
$1s^2$ $\boxed{2s^2\ 2p^2}$

valence electrons

Electronic Configurations

ex: Fe - atomic # = 26 $1s^2$ $2s^2$ $2p^6$ $3s^2$ $3p^6$ $\boxed{3d^6 \ 4s^2}$

valence electrons



N-shell $n = 4$

M-shell $n = 3$

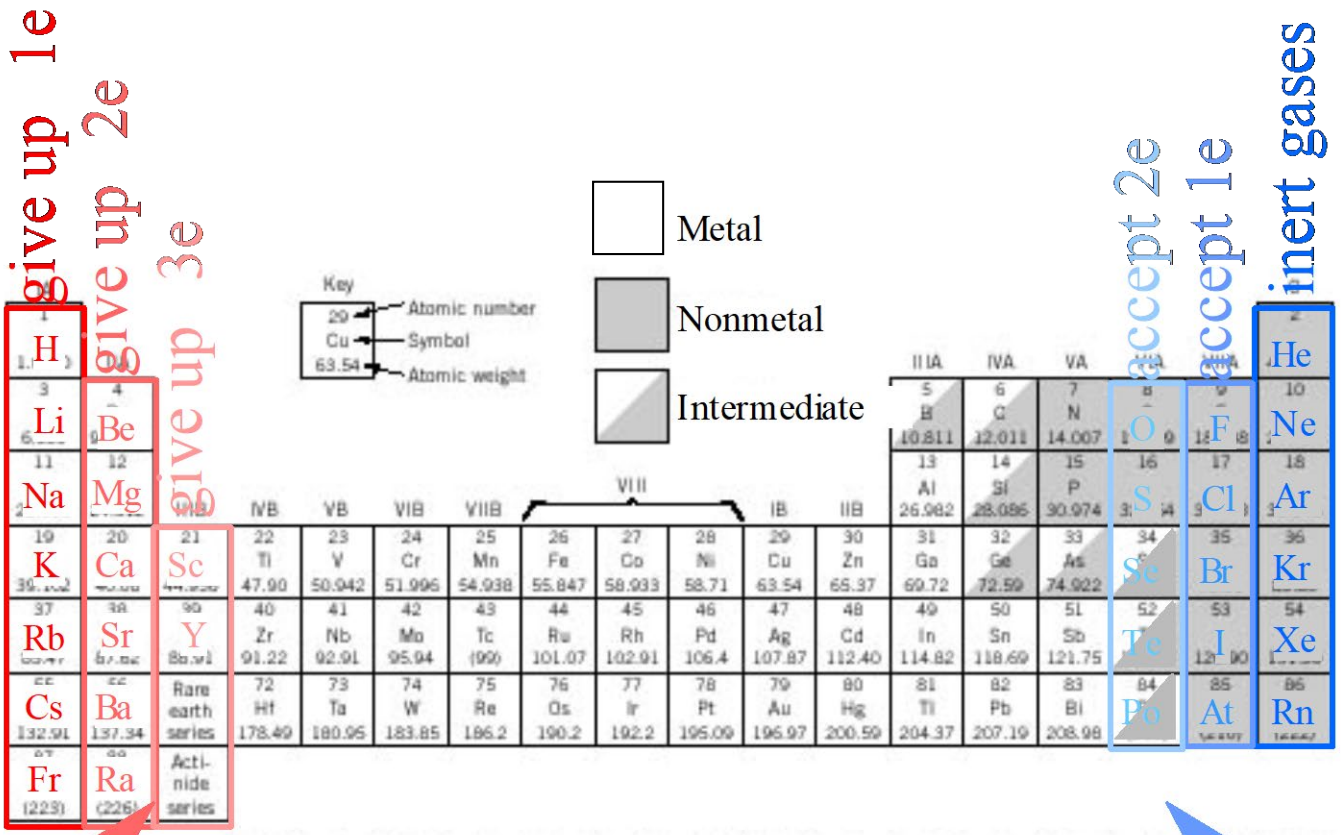
L-shell $n = 2$

K-shell $n = 1$

Adapted from Fig. 2.4, *Callister & Rethwisch 8e.*

THE PERIODIC TABLE

- Columns: Similar Valence Structure



Adapted from Fig. 2.6, *Callister 6e.*

Electropositive elements: Readily give up electrons to become + ions.

Electronegative elements: Readily acquire electrons to become - ions.

ELECTRONEGATIVITY

- Ranges from 0.7 to 4.0,
- Large values: tendency to acquire electrons.

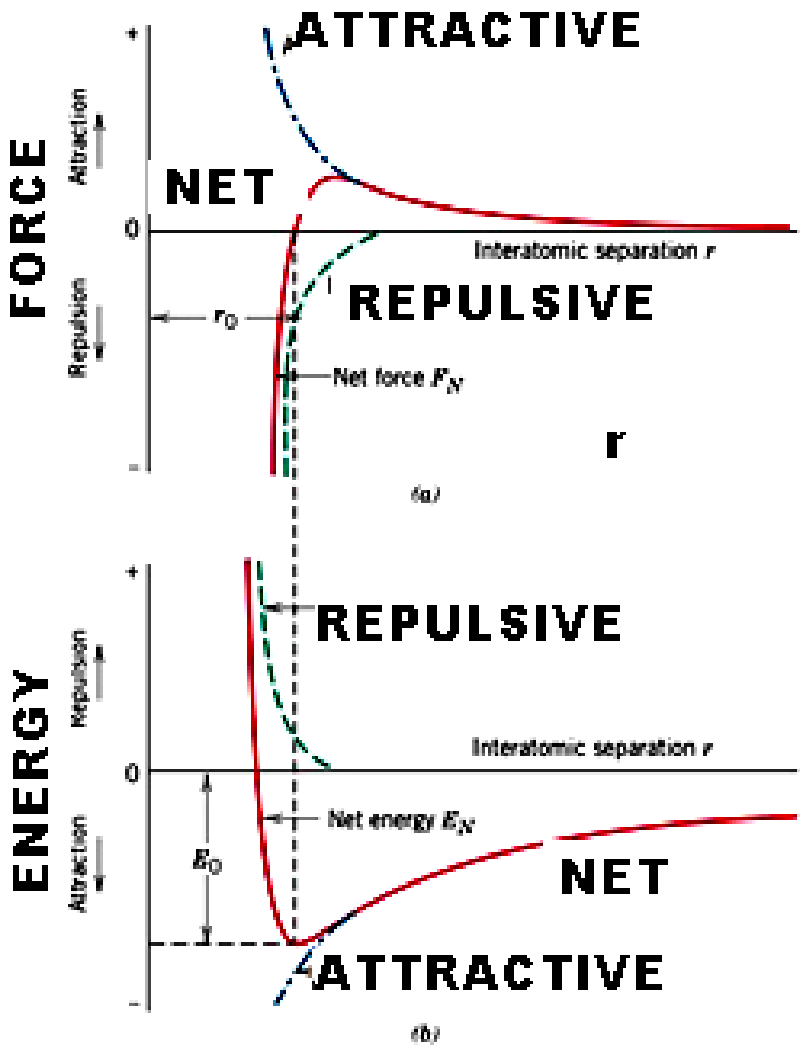
IA	IIA	IIIB	IVB	VB	VIB	VIIB	VIII			IB	IIB	IIIA	IVA	VA	VIA	VIIA	0
H 2.1																	He -
Li 1.0	Be 1.5											5 B 2.0	6 C 2.5	7 N 3.0	8 O 3.5	F 4.0	Ne -
Na 0.9	Mg 1.2											13 Al 1.5	14 Si 1.8	15 P 2.1	16 S 2.5	Cl 3.0	Ar -
K 0.8	Ca 1.0	21 Sc 1.3	Ti 1.5	23 V 1.6	Cr 1.6	25 Mn 1.5	Fe 1.8	27 Co 1.8	Ni 1.8	29 Cu 1.9	Zn 1.8	31 Ga 1.6	32 Ge 1.8	As 2.0	34 Se 2.4	Br 2.8	Kr -
Rb 0.8	Sr 1.0	39 Y 1.2	40 Zr 1.4	41 Nb 1.6	42 Mo 1.8	43 Tc 1.9	44 Ru 2.2	45 Rh 2.2	46 Pd 2.2	47 Ag 1.9	48 Cd 1.7	49 In 1.7	50 Sn 1.8	51 Sb 1.9	52 Te 2.1	I 2.5	Xe -
Cs 0.7	Ba 0.9	57–71 La–Lu 1.1–1.2	72 Hf 1.3	73 Ta 1.5	74 W 1.7	75 Re 1.9	76 Os 2.2	77 Ir 2.2	78 Pt 2.2	79 Au 2.4	80 Hg 1.9	81 Tl 1.8	82 Pb 1.8	83 Bi 1.9	84 Po 2.0	At 2.2	Rn -
Fr 0.7	Ra 0.9	89–103 Ac–No 1.1–1.7															

Smaller electronegativity

Larger electronegativity

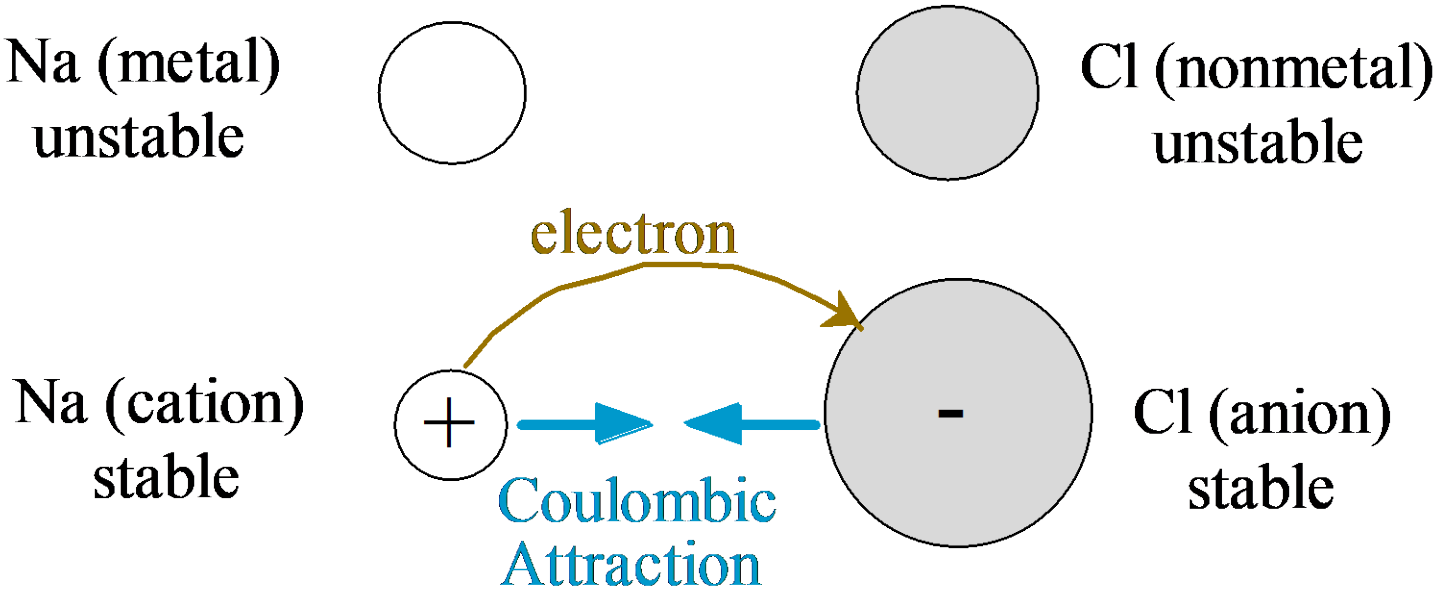
Adapted from Fig. 2.7, *Callister 6e.* (Fig. 2.7 is adapted from Linus Pauling, *The Nature of the Chemical Bond*, 3rd edition, Copyright 1939 and 1940, 3rd edition. Copyright 1960 by Cornell University.

FORCES AND ENERGIES



IONIC BONDING

- Occurs between + and - ions.
- Requires electron transfer.
- Large difference in electronegativity required.
- Example: NaCl



Ionic bond – metal + nonmetal

↑ donates electrons

↑ accepts electrons

Dissimilar electronegativities

ex: MgO

Mg $1s^2\ 2s^2\ 2p^6\ 3s^2$

[Ne] $3s^2$

O $1s^2\ 2s^2\ 2p^4$

Mg^{2+} $1s^2\ 2s^2\ 2p^6$

[Ne]

O^{2-} $1s^2\ 2s^2\ 2p^6$

[Ne]

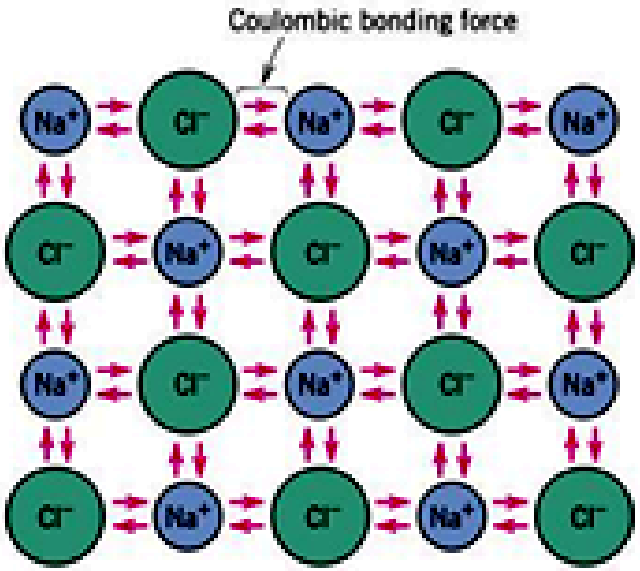


Figure 2.9
bonding in s

FORCES, ATTRACTION F_A

Bonding Forces and Energies

2.13 ***Calculate the force of attraction between a K^+ and an O^{2-} ion the centers of which are separated by a distance of r0 =1.5 nm.***

<u>Solution</u>

The attractive force between two ions *FA* is just the derivative with respect to the interatomic separation of the attractive energy expression, Equation 2.8, which is just

$$F_A = \frac{dE_A}{dr} = \frac{d\left(-\frac{A}{r}\right)}{dr} = \frac{A}{r^2}$$

The constant *A* in this expression is defined in footnote 3. Since the valences of the K+ and O2- ions

(*Z*1 and *Z*2) are +1 and -2, respectively, *Z*1 = 1 and *Z*2 = 2, then

$$F_A = \frac{(Z_1 e)(Z_2 e)}{4\pi\varepsilon_0 r^2}$$

$$= \frac{(1)(2)(1.602 \times 10^{-19}\ \mathrm{C})^2}{(4)(\pi)(8.85 \times 10^{-12}\ \mathrm{F/m})(1.5 \times 10^{-9}\ \mathrm{m})^2}$$

=2.05 × 10^(-10) N

IONIC FORCE / P 31 FOOT-NOTE

$F = (Z1 * Z2 * e^2)/(4*\pi*\varepsilon_0*r^2)$;

$e = 1.602 * 10^{(-19)}$ COULOMBS ;

$\varepsilon_0 = 8.85 * 10^{(-12)}$

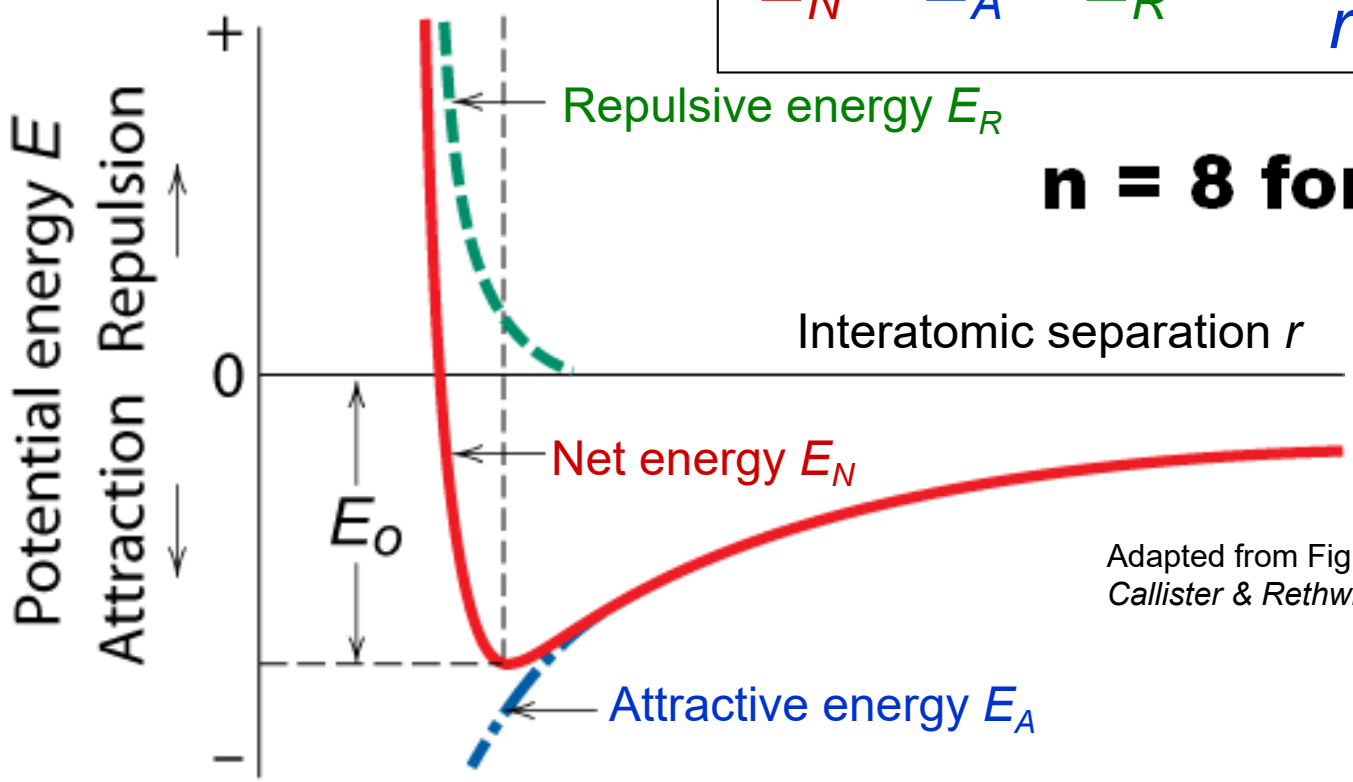
Z1, Z2 = VALENCIES OF IONS

Ionic Bonding

- Energy – minimum energy most stable
  - Energy balance of attractive and repulsive terms

$$E_N = E_A + E_R = -\frac{A}{r} + \frac{B}{r^n}$$

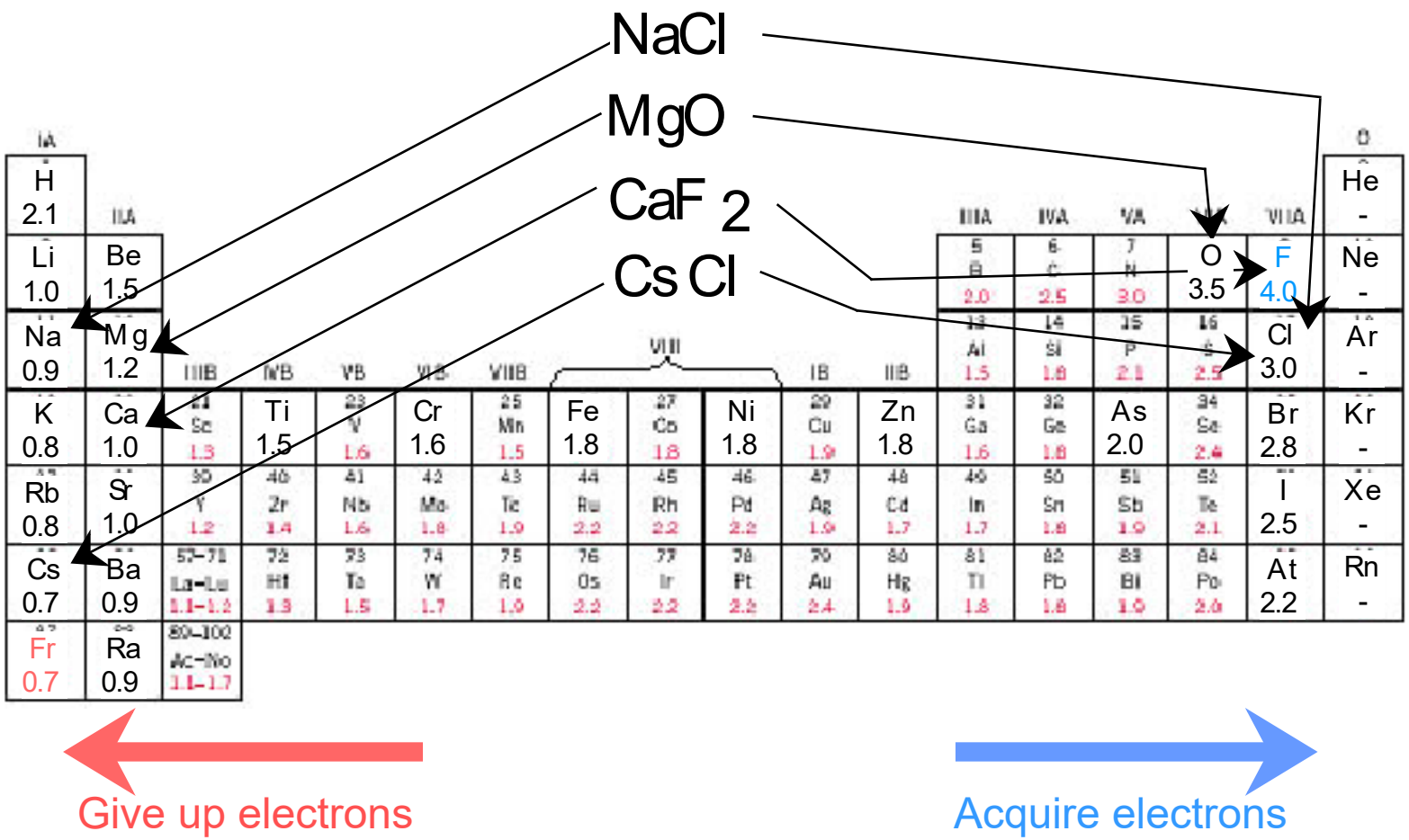
n = 8 for ions



Adapted from Fig. 2.8(b), *Callister & Rethwisch 8e.*

EXAMPLES: IONIC BONDING

- Predominant bonding in Ceramics



Adapted from Fig. 2.7, *Callister 6e.* (Fig. 2.7 is adapted from Linus Pauling, *The Nature of the Chemical Bond*, 3rd edition, Copyright 1939 and 1940, 3rd edition. Copyright 1960 by Cornell University.

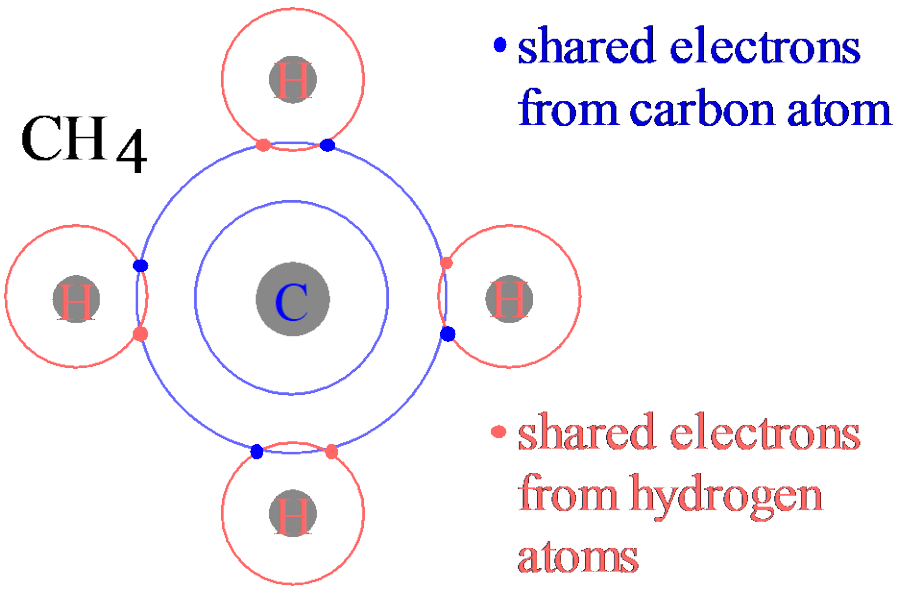
COVALENT BONDING

- Requires shared electrons
- Example: CH_4

C: has 4 valence e, needs 4 more

H: has 1 valence e, needs 1 more

Electronegativities are comparable.



Adapted from Fig. 2.10, *Callister 6e.*

Primary Bonding

- Metallic Bond -- delocalized as electron cloud

- Ionic-Covalent Mixed Bonding

$$\% \text{ ionic character} = \left(1 - e^{-\frac{(X_A - X_B)^2}{4}}\right) \text{x}\,(100\%)$$

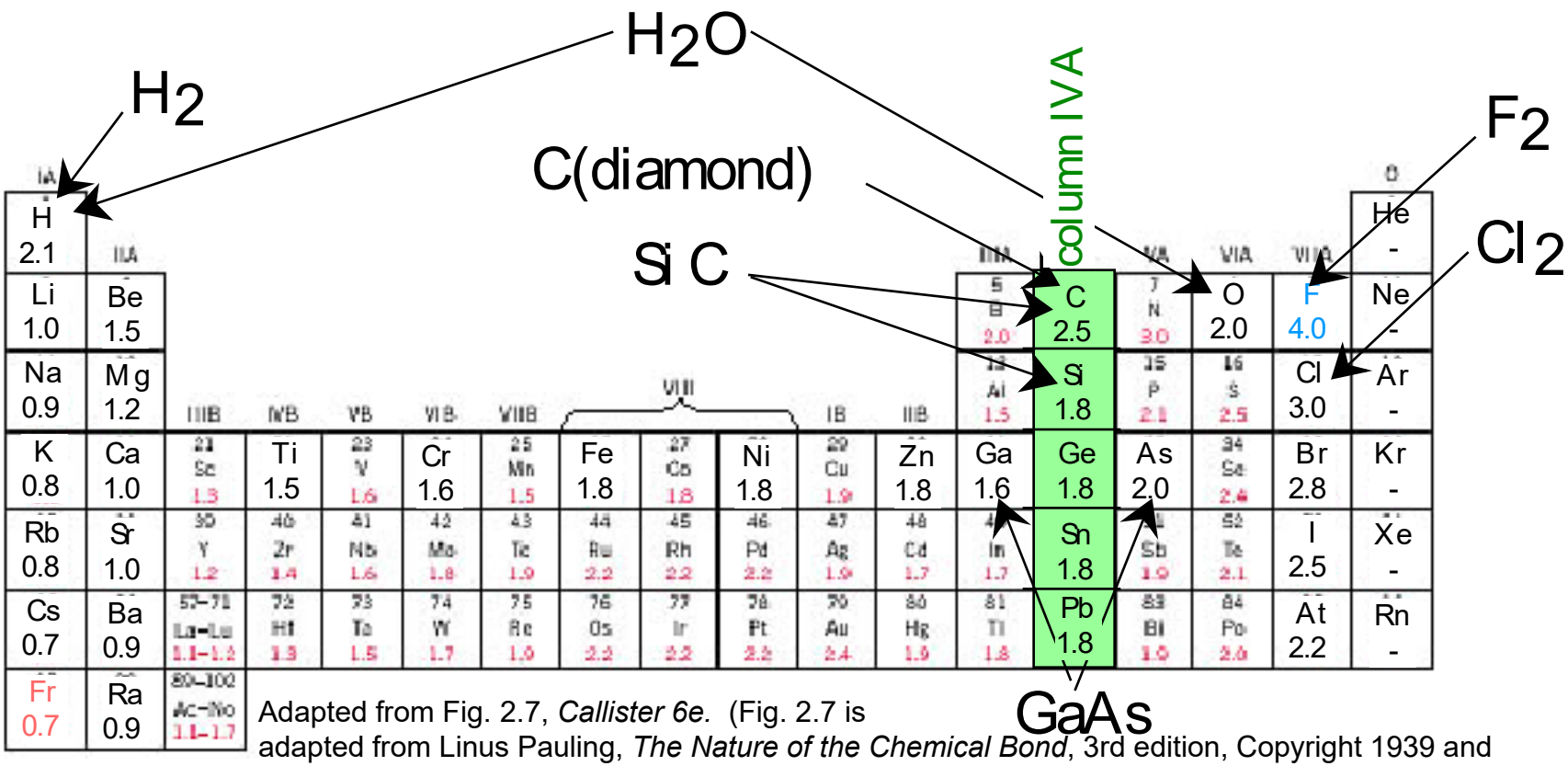
where X_A & X_B are Pauling electronegativities

Ex: MgO

$X_{Mg} = 1.2$
$X_O = 3.5$

$$\% \text{ ionic character} = \left(1 - e^{-\frac{(3.5-1.2)^2}{4}}\right) \text{x}\,(100\%) = 73.4\% \text{ ionic}$$

EXAMPLES: COVALENT BONDING

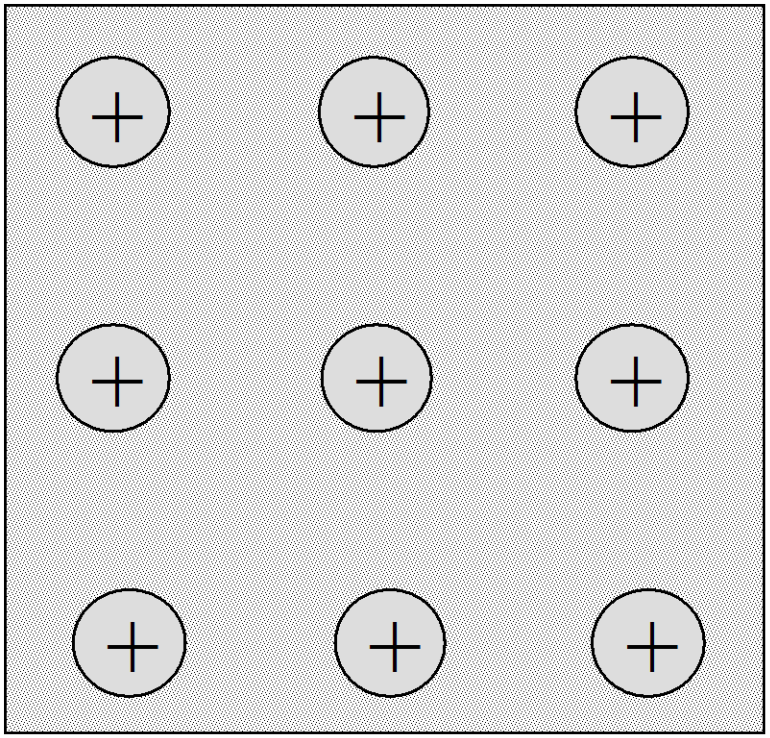


Adapted from Fig. 2.7, *Callister 6e.* (Fig. 2.7 is adapted from Linus Pauling, *The Nature of the Chemical Bond*, 3rd edition, Copyright 1939 and 1940, 3rd edition. Copyright 1960 by Cornell University.

- Molecules with nonmetals
- Molecules with metals and nonmetals
- Elemental solids (RHS of Periodic Table)
- Compound solids (about column IVA)

METALLIC BONDING

- Arises from a sea of donated valence electrons (1, 2, or 3 from each atom).



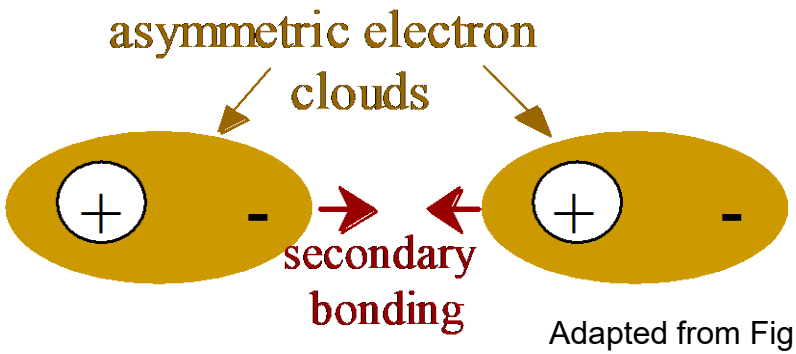
Adapted from Fig. 2.11, *Callister 6e.*

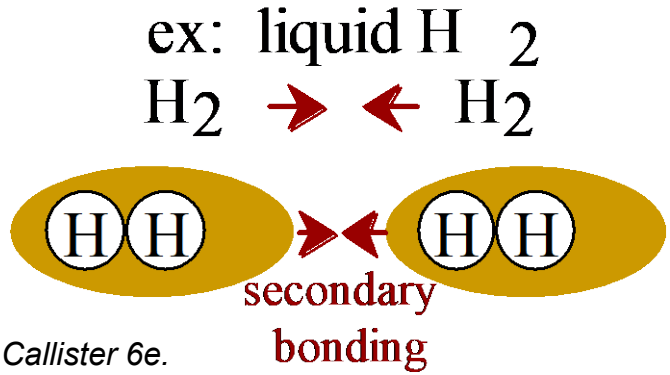
- Primary bond for metals and their alloys

SECONDARY BONDING

Arises from interaction between dipoles

- Fluctuating dipoles





Adapted from Fig. 2.13, *Callister 6e.*

- Permanent dipoles-molecule induced

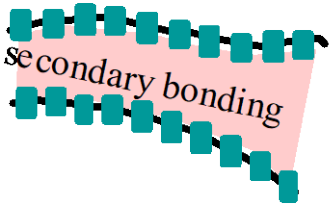
-general case:

Adapted from Fig. 2.14, *Callister 6e.*

-ex: liquid HCl

Adapted from Fig. 2.14, *Callister 6e.*

-ex: polymer

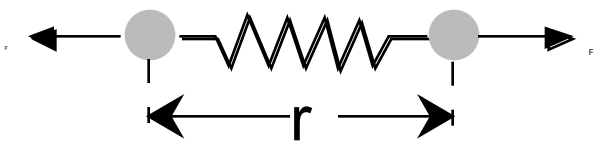


SUMMARY: BONDING

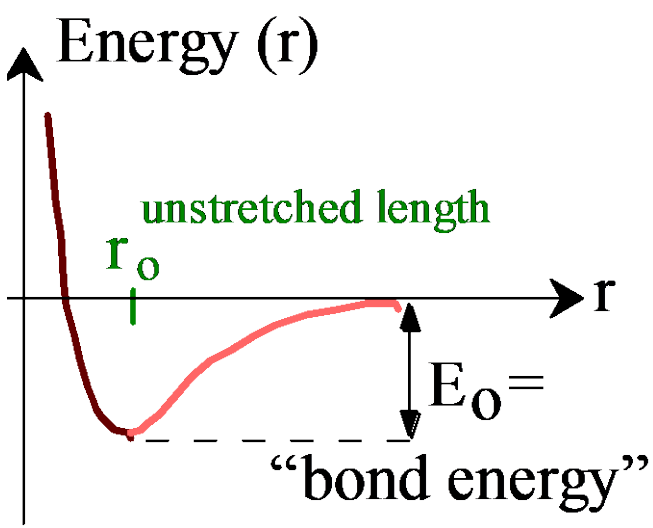
Type	Bond Energy	Comments
Ionic	Large!	Nondirectional (ceramics)
Covalent	Variable large-Diamond small-Bismuth	Directional semiconductors, ceramics polymer chains)
Metallic	Variable large-Tungsten small-Mercury	Nondirectional (metals)
Secondary	smallest	Directional inter-chain (polymer) inter-molecular

PROPERTIES FROM BONDING: T_M

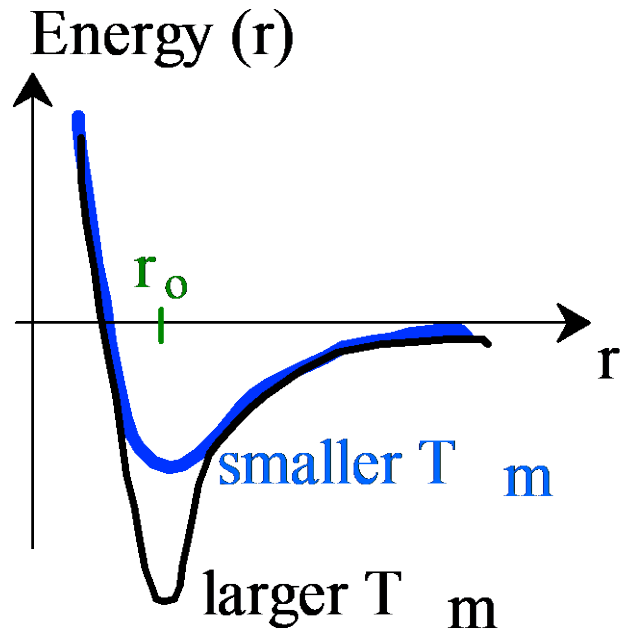
- Bond length, r



- Bond energy, E_o



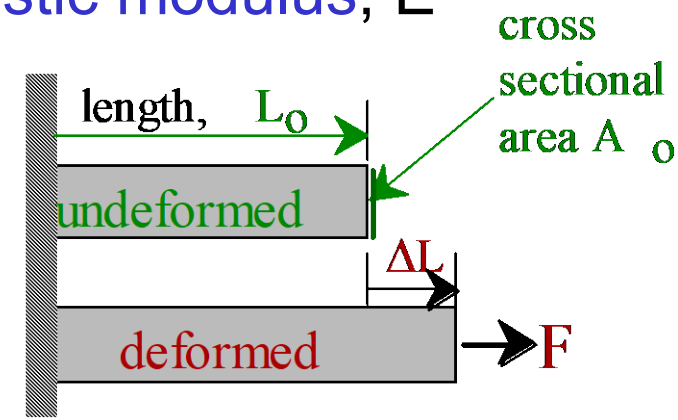
- Melting Temperature, T_m



T_m is larger if E_o is larger.

PROPERTIES FROM BONDING: E

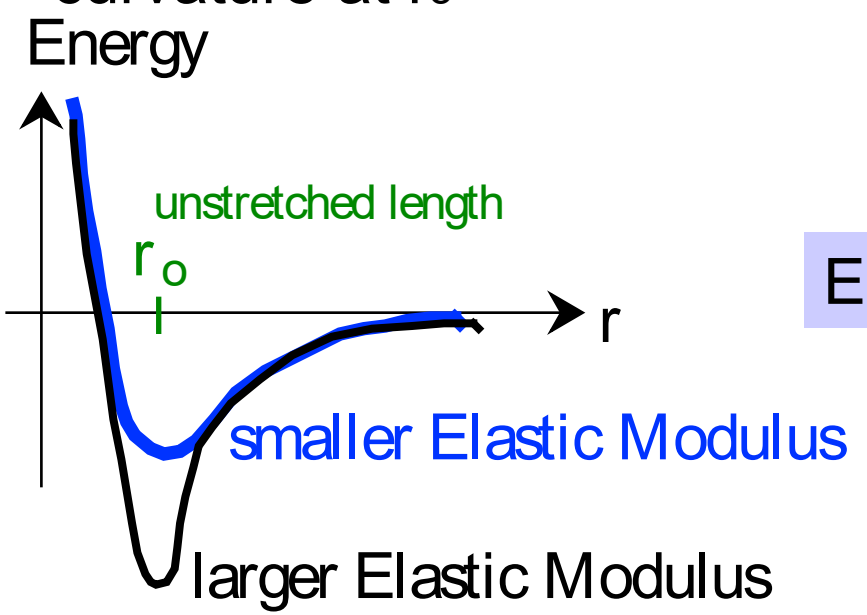
- Elastic modulus, E



Elastic modulus

$$\frac{F}{A_o} = E \frac{\Delta L}{L_o}$$

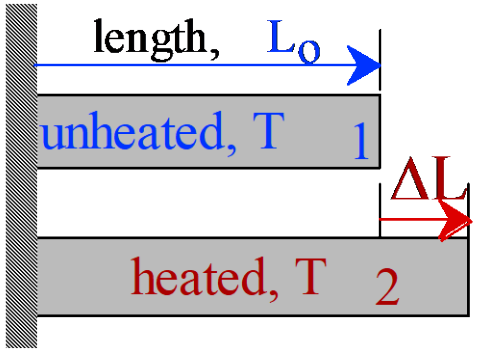
- E ~ curvature at r_o



E is larger if E_o is larger.

PROPERTIES FROM BONDING: α

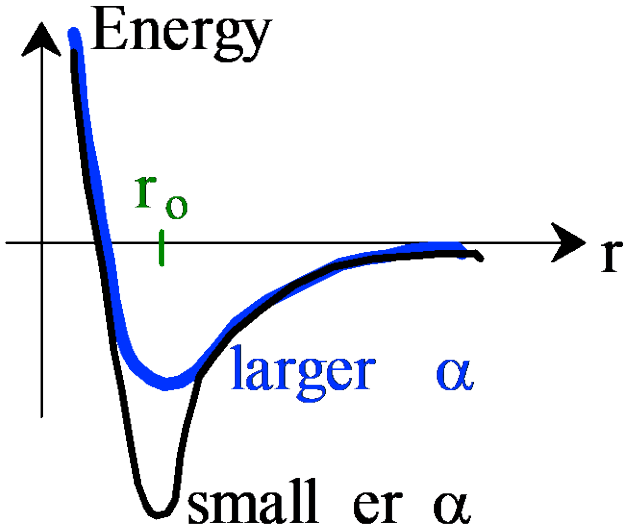
- Coefficient of thermal expansion, α



coeff. thermal expansion

$$\frac{\Delta L}{L_o} = \alpha (T_2 - T_1)$$

- α ~ symmetry at r_o



α is larger if E_o is smaller.

SUMMARY: PRIMARY BONDS

Ceramics

(Ionic & covalent bonding):

Large bond energy

large T_m
large E
small α

Metals

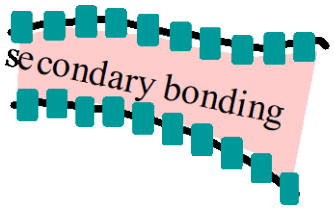
(Metallic bonding):

Variable bond energy

moderate T_m
moderate E
moderate α

Polymers

(Covalent & Secondary):



Directional Properties

Secondary bonding dominates
small T
small E
large α